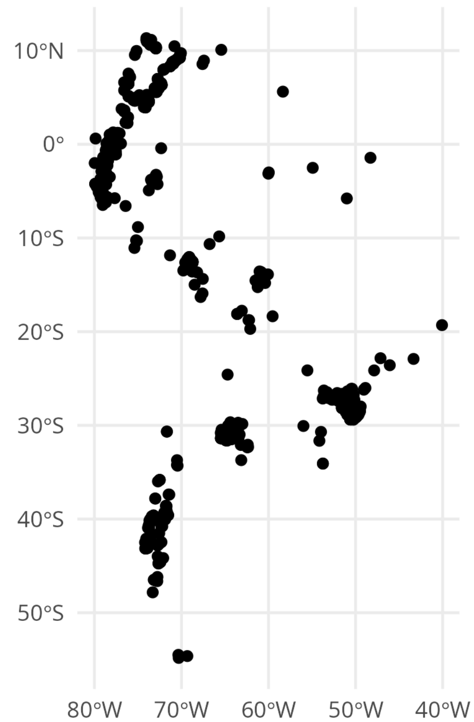
STeMP: Spatio-TEmporal Modelling Protocol

Table

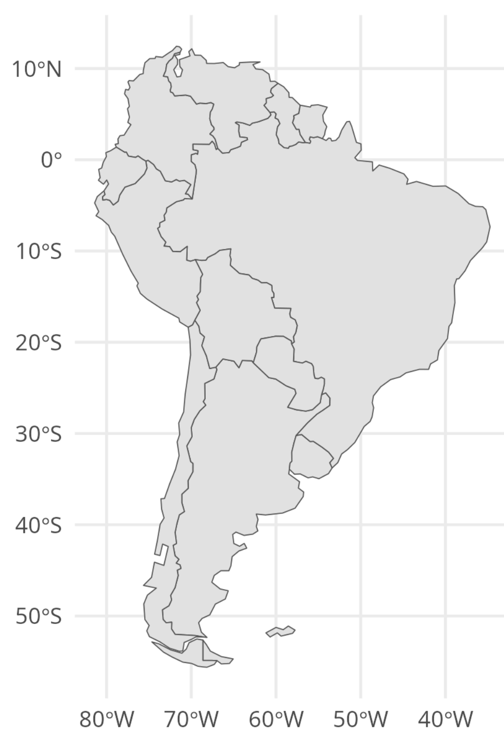
section	element	value
Overview	Model title	Spatial predictive modelling of plant species richness in South America
Overview	Author names	Anonymous
Overview	Contact	anonymous@xy.com
Overview	Study title	
Overview	Study link	
Overview	Target variable	plant species richness
Overview	Scientific Field	Biogeosciences
Model	Model type	Regression
Model	Learning method	ranger
Model	Architecture	
Model	Training domain	regional
Model	Sample acquisition	In-Situ
Model	Sample geometry	Point
Model	Sample size	703
Model	Range	1 to 256
Model	Sampling design	clustered
Model	Coordinate Reference System	EPSG:4326
Model	Data sources of response	sPlotOpen
Model	Predictor types	Remote Sensing Images, Spatial Proxies
Model	Number of predictors	6
Model	Names of predictors	bio_4, bio_12, bio_8, elev, X, Y
Model	Preprocessing	Resampling
Model	Data sources of predictors	WorldClim, SRTM
Model	Validation strategy	Random Cross-Validation
Model	Performance metrics	Rsquared, RMSE
Model	Validation results	RMSE = 22.330, R^2 = 0.737

section	element	value
Model	Hyperparameter tuning	mtry=6, splitrule=1, min.node.size=5
Model	Predictor selection	None
Model	Explainability	
Model	Scientific interpretation	
Model	Potential biases	Training points highly clustered, large parts of South America not covered, important predictors likely missing and instead spatial proxies were used
Model	Sensitivity assessment	
Model	Limitations	Extrapolation likely required for large parts of South America, Model not tested for its ability to make predictions to new regions
Model	Software	R
Model	Code availability	Zenodo
Model	Data availability	Zenodo
Prediction	Prediction domain	regional
Prediction	Evaluation strategy	Random Cross-Validation
Prediction	Performance metrics	Rsquared
Prediction	Map accuracy	R^2 = 0.737
Prediction	Uncertainty quantification	None
Prediction	Threshold selection	
Prediction	Post-processing	

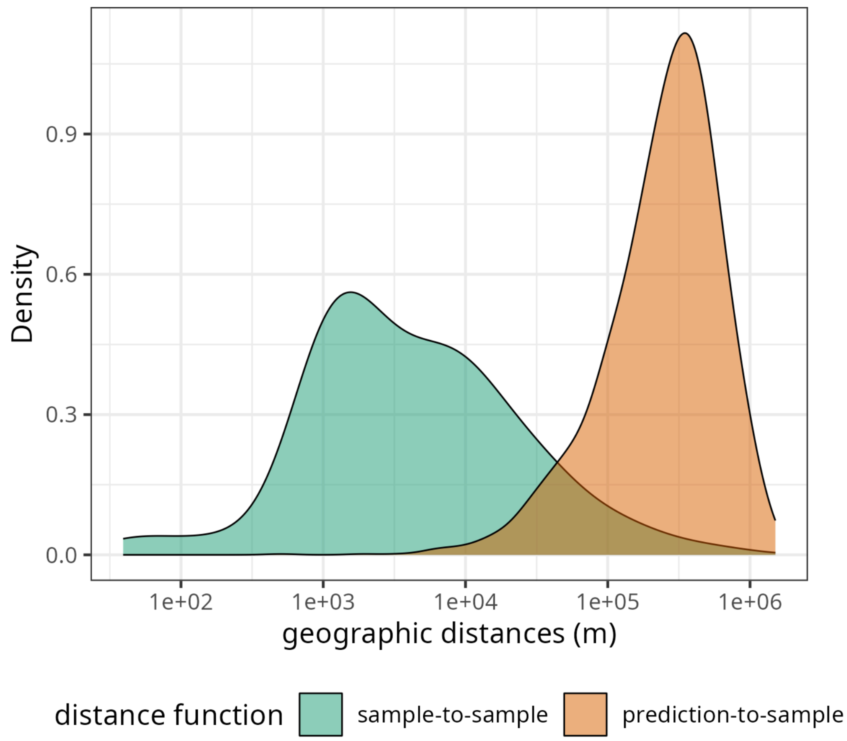
Figures



Map of sampling locations.



Map of prediction area.



Plot of nearest neighbour distances in geographic space.